THE IMAGE OF AN IRREDUCIBLE SUBSET UNDER A CONTINUOUS MAP IS IRREDUCIBLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-topology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-topology/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

Editorial note. Bibliography provenance: neither the counted statement nor its proof contains a citation (this lane rescans the whole span before writing the card), so the card carries no bibliography and no entry was reconstructed.

Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Lemma 0.1. *Let $f : X \to Y$ be a continuous map of topological spaces. If $E \subset X$ is an irreducible subset, then $f(E) \subset Y$ is irreducible as well.*

Proof. Clearly we may assume $E = X$ (i.e., X irreducible) and $f(E) = Y$ (i.e., f surjective). First, $Y \neq \emptyset$ since $X \neq \emptyset$. Next, assume $Y = Y_1 \cup Y_2$ with Y_1, Y_2 closed. Then $X = X_1 \cup X_2$ with $X_i = f^{-1}(Y_i)$ closed in X. By assumption on X, we must have $X = X_1$ or $X = X_2$, hence $Y = Y_1$ or $Y = Y_2$ since f is surjective. □

Date: stacks-project:topology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.